ritmoForceCurve

Oar Sensor Kit Instructions

Separate setup and recording instructions for iPhone/iOS and Android

Before you launch: charge the phone and both oar sensors.

Keep track of which physical sensor is the INNER/SHAFT sensor and which is the OUTER/BLADE sensor.

Do the alignment/calibration movements carefully. Good setup matters more than doing them quickly.

Website: www.ritmoForceCurve.com

These instructions are intended to be packed with the sensor kit. App screens may move slightly as software is updated; the workflow remains the same.

PART 1 - iPHONE / iOS

Use the RitmoForceCurve iPhone app for sensor connection, oar setup, live rowing display, workout recording, phone motion/GPS collection, and cloud upload.

1. Charge, identify, and mount the two sensors

1. Charge both WitMotion sensors and your iPhone before going on the water.
2. Identify the two physical sensors. Keep the same identity every session: INNER/SHAFT = the sensor closer to the handle/pin; OUTER/BLADE = the sensor farther out toward the blade.
3. Install each sensor in its supplied housing/saddle and fasten it firmly to the oar. The sensor must not slide, rotate, or rock during rowing.
4. Use the marked kit positions when provided. Typical two-sensor geometry is approximately 10 in (25 cm) and 45 in (114 cm) from the pin, but use the actual saved/measured dimensions for your oar.
5. Mount both sensors in the same repeatable orientation. The button/USB-C orientation should match the kit markings. Do not casually reverse a sensor between sessions.

Mounting check: A sensor that moves on the shaft can look like oar bend. Tight mounting is essential.

2. Power on and connect in RitmoForceCurve

6. Turn on both sensors before starting the session.
7. Open RitmoForceCurve on the iPhone.
8. Open Connect Oar Sensors. Scan for nearby sensors and select the two units belonging to your oar.
9. Confirm that the correct physical sensor is assigned to the inner/shaft position and the outer/blade position. Saved oars can be selected for faster reconnection.
10. Verify that both sensors are connected and live data are changing before continuing.

Do not use the iPhone Bluetooth Settings page to manage the rowing connection. Connect the sensors from inside RitmoForceCurve.

3. Select or create the oar

11. Choose your saved oar, or create a new oar if this is the first use.
12. Enter/confirm the oar type and side (Port or Starboard for sweep; the correct configuration for sculling).
13. Enter the measured oar dimensions requested by the app, including the sensor positions and inboard/outboard dimensions. These values are saved with the oar and uploaded with workouts.
14. Use the saved bend/force scale factor for that oar. Only override it when you intentionally want to change the force calibration.

4. Run oar alignment / calibration

Follow the prompts in the app. Perform the movements smoothly and deliberately. The current iOS sequence uses six steps; the last two are optional water-reference movements.

1 - Bias / still: Hold the oar completely still in the prompted reference position while the app measures stationary sensor bias.

2 - Feather and square: With the shaft otherwise steady, rotate the blade repeatedly between fully feathered and fully squared. Use clean, full rotations rather than small wrist movements.

3 - Feathered level sweep: Keep the blade feathered and sweep the oar through a normal rowing arc while keeping the shaft approximately level.

4 - Square level sweep: Keep the blade squared and sweep through a normal rowing arc while keeping the shaft approximately level.

5 - Square swoop - optional: With the blade squared, make the prompted smooth swooping motion. This supplies an additional 3-D alignment reference. Skip only when the app allows it and conditions make it impractical.

6 - Squared float - optional: Square the blade, let it float naturally on the water, and keep the oar steady for about 12 seconds. The app uses the stable middle portion of the hold as a water-reference measurement.

Floating square blade: Do not force the blade deeper or hold it artificially high. Let the squared blade find its natural floating position.

5. Final pre-row checks

- Confirm both sensors remain connected.
- Confirm the correct oar, side, and calibration are active.
- If the app asks for water depth in Workout Settings, enter the current value before the workout. It is stored separately from the alignment data.
- Confirm GPS/location permission is available to the app.
- Secure the iPhone so it cannot move independently of the boat if you are using phone motion for boat-motion analysis.

6. Record the workout

15. From the workout screen, tap Start Session / Start Workout.
16. Row normally. The iPhone app records the oar sensors together with phone motion, GPS, timing, and the active oar/calibration settings.
17. Use the live display as desired. Do not change sensor mounting during the session.
18. At the end of rowing, tap Stop Session. Allow the app to finish saving before turning off the sensors.

If a sensor disconnects: Reconnect when practical. Do not swap the two physical sensors or change their mounting orientation mid-session.

7. Upload for Cloud Analysis

19. Open the completed workout / Cloud Analysis area in RitmoForceCurve.
20. Upload the workout. The app packages the sensor recordings, phone motion, GPS, workout metadata, oar dimensions, and calibration/alignment information needed by the server.
21. Keep the phone connected long enough for the upload to complete. If upload is interrupted, use the retry/upload option rather than making a second unrelated workout.
22. Open the completed report from the app or website when processing is finished.

8. iOS quick troubleshooting

Problem	What to check
Only one sensor appears	Make sure both sensors are powered on. Rescan in RitmoForceCurve and verify you selected the correct saved sensor addresses.
Curves look reversed or implausible	Check oar side and confirm inner/shaft vs outer/blade sensor assignment. Do not remount one sensor backwards.
Force/bend looks noisy	Check that both housings are tight and cannot twist or slide on the shaft.
GPS/boat motion missing	Check iPhone location/motion permissions and make sure the phone was fixed to the boat rather than carried loosely.
Alignment looks poor	Repeat the alignment carefully. Full clean motions are better than fast or exaggerated motions.

iOS KIT / OAR RECORD

Fill this in once for each kit/oar. It makes sensor identification and troubleshooting much easier.

Oar name / boat	
INNER/SHAFT sensor address	
OUTER/BLADE sensor address	
Inner sensor position	
Outer sensor position	
Bend/force scale factor / notes	

PART 2 - ANDROID

Android currently uses the WitMotion Android app for the two oar sensors and Sensor Logger for the phone motion/GPS recording. The files are then uploaded together for RitmoForceCurve Cloud Analysis.

1. Install the recording apps

- WitMotion Android app - used to connect to and record both oar sensors.
- Sensor Logger - used in the background to record the Android phone accelerometer/gyroscope and GPS at their appropriate rates.

Important: Pair/connect the WitMotion sensors inside the WitMotion app - not from the Android system Bluetooth Settings screen.

2. Charge, identify, and mount the two sensors

23. Charge both WitMotion sensors and the Android phone.
24. Identify the sensors and keep their identities consistent: INNER/SHAFT is closer to the handle/pin; OUTER/BLADE is farther toward the blade.
25. Install each sensor in the supplied housing/saddle and tighten it firmly so it cannot slide or rotate.
26. Use the marked kit positions when supplied. If dimensions are entered during upload/setup, use the actual measured distances rather than estimates.
27. Keep sensor orientation repeatable from one session to the next.

3. Connect both oar sensors in WitMotion

28. Turn on both sensors.
29. Open the WitMotion app. Allow the requested Bluetooth/nearby-device/location permissions required by your Android version.
30. From the main screen, open the connection/device list (chain/link icon on versions that use it).
31. Select the correct product/device type and switch to the multiple-device mode (shown as “Multip” on current versions).
32. Connect both sensors by their name/address. If other crews are nearby, use the recorded addresses rather than guessing from signal strength.
33. Verify that both devices show changing live values before recording.

4. Record the five Android alignment movements

Make one dedicated alignment recording with the five blocks below, in order. Except for the first stationary block, the moving blocks do not need extra stillness before or after each movement.

Movement 1 - Bias / still: Hold the oar and both sensors completely still long enough to establish stationary bias.

Movement 2 - Feather and square: Keep the shaft otherwise steady and rotate the blade repeatedly between feathered and fully squared.

Movement 3 - Feathered level sweep: Keep the blade feathered and sweep through a normal rowing arc with the shaft approximately level.

Movement 4 - Square level sweep: Keep the blade squared and sweep through a normal rowing arc with the shaft approximately level.

Movement 5 - Floating square blade: With the blade fully squared, place it on the water and let it float naturally. Keep the oar steady for about 12 seconds. Do not force the blade to an artificial depth.

Movement 5 is the water reference. The stable middle portion of the floating hold is used; avoid bumping the oarlock or pushing down on the handle.

5. Prepare Sensor Logger on the Android phone

34. Open Sensor Logger and grant motion and location permissions. Allow background operation; on phones with aggressive battery saving, use unrestricted/no-battery-optimization for the recording period.
35. Record phone accelerometer and gyroscope at approximately 100 Hz.
36. Record GPS/location separately at about 1 Hz (1-2 Hz is appropriate). Do not create a combined file that repeats the same GPS value at the 100 Hz motion rate.
37. If orientation data are available in the normal Sensor Logger export, keep them; the importer can use the separate streams and their timestamps.

Phone placement: For boat-motion analysis, secure the Android phone rigidly to the boat. A phone moving in a pocket or bag records the phone, not the boat.

6. Record the rowing session

38. Start the Sensor Logger recording on the phone.
39. Return to the WitMotion app and start recording both oar sensors. Keep WitMotion in the foreground while rowing.
40. Row normally. Do not swap sensors or alter their mounting during the session.
41. At the end, stop the WitMotion recording, then stop Sensor Logger.
42. Before powering off the sensors, verify that the WitMotion recording has been saved.

7. Export the Android files

43. From WitMotion, retain/export the saved TXT recording (or the zipped recording folder if that is how your app version exports the session).
44. From Sensor Logger, choose the normal “CSV in Zip File” / zipped CSV export. Keep acceleration, gyroscope, location/GPS, and orientation as separate files in the ZIP.
45. Do not convert everything into one combined CSV. Different streams have different sample rates and timestamps.
46. Keep the alignment recording separate and clearly identifiable if it is not already associated with the oar in the upload workflow.

8. Upload for RitmoForceCurve Cloud Analysis

47. Open www.ritmoForceCurve.com and go to Cloud Analysis / Android upload.
48. Upload the WitMotion sensor recording together with the Sensor Logger ZIP for the same rowing session.
49. Provide/confirm the oar identity, side, dimensions, sensor assignment, and calibration values requested by the upload page.
50. When an alignment file is requested, supply the matching five-movement alignment recording for that oar/sensor mounting.
51. Submit the workout and open the report when processing completes.

9. Android quick troubleshooting

Problem	What to check
Sensors visible in Android Settings but not WitMotion	Forget/ignore system pairing and connect them from inside the WitMotion app.
Only one sensor connects	Confirm multiple-device/“Multip” mode and verify both sensor addresses.
Wrong sensor identity	Record which address is INNER/SHAFT and which is OUTER/BLADE. Correct assignment is important for bend/force calculations.
Sensor Logger stops in background	Allow background operation and disable battery optimization for the recording period.
Huge or awkward phone file	Use separate Sensor Logger CSV streams. Keep GPS around 1-2 Hz rather than repeating it at motion-sensor rate.
Report says alignment is missing/weak	Use the matching alignment recording and repeat all five movements cleanly, especially the 12-second floating squared-blade hold.

PART 3 - AFTER UPLOAD: WHAT TO EXPECT

The cloud report combines the oar sensors with the available phone/GPS data and the saved oar/calibration information. Depending on the data recorded, reports can include stroke timing, rate, bend/force, work/power, oar motion, catch/finish technique metrics, speed, GPS/boat-motion information, and grouped/average force curves.

- Do not add together metrics that describe a per-stroke “check” or variation. Values such as speed check, distance check, and energy-loss check are summarized by averaging across strokes/groups/pieces/workouts rather than summing them.
- Keep original recordings until you have opened and checked the finished report.
- If a result looks physically impossible, first verify sensor identity, mounting, oar side, alignment, dimensions, and phone mounting before changing calibration factors.

FAST PRE-LAUNCH CHECK

Check	iOS	Android
Both oar sensors charged/on	Yes	Yes
Correct inner/outer sensor identity	Yes	Yes
Sensors tight and orientation unchanged	Yes	Yes
Oar dimensions/calibration available	Saved in app	Confirm for upload
Alignment completed	In RitmoForceCurve	5-movement WitMotion recording
Phone fixed to boat for motion analysis	Yes	Yes
GPS/location enabled	RitmoForceCurve	Sensor Logger
Phone IMU recording	Automatic in app	Sensor Logger ~100 Hz
Oar sensor recording	Automatic in app	WitMotion foreground
Upload	From app	WitMotion + Sensor Logger files

Support / updates: www.ritmoForceCurve.com

Keep this sheet with the kit so the same sensor identities, mounting orientation, and alignment procedure are used every session.